6 A sprinter takes a time of 11.0 s to run a 100 m race. She first accelerates uniformly from rest, reaching a speed of $10\,m\,s^{-1}$. She then runs at a constant speed of $10\,m\,s^{-1}$ until the finish line.

What is the uniform acceleration of the sprinter for the first part of the race?

A $0.5\,m\,s^{-2}$ **B** $0.91\,m\,s^{-2}$ **C** $1.7\,m\,s^{-2}$ **D** $5.0\,m\,s^{-2}$